CHICAGO CRIME ANALYSIS & PREDICTION (2020–2022)

Synthesizing Machine Learning Models & Power BI Analytics for Public Safety

A comprehensive data science study evaluating ~500K records to uncover spatial hotspots, temporal dynamics, arrest predictors, and domestic violence risk factors.

Prepared by: Tan Kai Boon | GitHub: tKaiB/Capstone_Chicago

Project Scope: Chicago Crime

Context

The Chicago Crime Analytics dataset contains crime incident records reported within the City of Chicago.

The dataset is designed to *support public sector analytics, crime trend analysis, law enforcement reporting, geographic crime mapping, and community safety initiatives*.

By combining crime incidents with crime classifications and location categories, analysts can identify crime patterns, evaluate public safety performance, analyze geographic hotspots, and support evidence-based policy decisions.

Core Research Questions

1. Spatial Hotspots

Which Chicago community areas or districts exhibit the highest crime concentrations, and do these hotspots remain spatially stable across 2020–2022?

2. Temporal Patterns

How do crime frequencies vary across day, month, and season, and are there identifiable cyclical patterns that persist year-over-year?

3. Arrest & Clearance Dynamics

What crime types and locations drive lower arrest rates, and can machine learning predict arrest likelihood accurately?

4. Crime Type Dynamics

How has the composition of crime types (violent vs. property vs. other) shifted between 2020 and 2022, and which categories show the most volatility?

5. Domestic & Location Risk

How do domestic-flagged incidents and location descriptors (street, residence, commercial) vary across community areas, and did the pandemic year of 2020 meaningfully alter these distributions?

Project Scope: Dataset Context, Schema & Timeline

Schema (Data Dictionary)

Tables
Crime Types, Location Types, Crime 2020/21/22

Key Columns
Arrest (Boolean), Domestic (Boolean), Crime_Type_ID, Location_Type_ID, Community Area, Latitude/Longitude

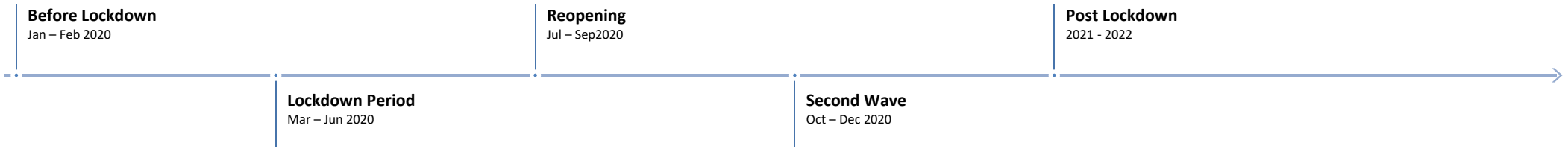
Dropped Columns
District, Ward, Beat (collinear)

Dataset Scope & Pandemic Context

Dataset Size & Filtering
639,406 raw incidents condensed to 627,163 cleaned records (2020-2022) after removing records lacking geographic coordinates or ward assignments.

Why 2020–2022 Matters
This timeframe captures the full arc of the COVID-19 pandemic, from pre-lockdown baseline through acute lockdown, reopening, second wave, and post-lockdown recovery.

Multi-Source Synthesis
Integrates raw Chicago Police Department incident logs, engineered temporal/spatial features, Power BI dashboards, and XGBoost machine learning pipelines.



Crime Patterns at a Glance

TOTAL INCIDENTS
627,163
Cleaned records (2020-2022)

ARREST RATE
13.25%
83K arrests / 544K non-arrests

DOMESTIC RATE
19.72%
124K domestic-flagged crimes

PROPERTY SHARE
49.69%
Dominant category overall

Temporal Heatmap & Weekly Incident Volume (Power BI Insights)

- **Friday Incident Peak:** Friday recorded the highest total incidents overall (91,886), followed closely by Saturday (91,533) and Sunday (90,655). Weekdays stay around 86K–89K.
- **Time of Day Concentration:** Afternoon (12:00–17:59) leads all time periods with 195,128 incidents, followed by Evening (18:00-23:59) at 181,953 incidents. Night (00:00–05:59) accounts for 121,141 incidents.
- **Seasonal Cyclicality:** Crime peaks consistently during Summer months (June–August) across all three years, dropping significantly during Winter (December–February) due to severe weather constraints.

Day_of_Week	Afternoon	Evening	Morning	Night	Total
Friday	29171	27423	19077	16215	91886
Saturday	26545	27566	16961	20461	91533
Sunday	25525	26102	16162	22866	90655
Monday	28366	25239	19349	16983	89937
Wednesday	29067	25713	19314	15013	89107
Tuesday	28396	25129	19332	14749	87606
Thursday	28058	24781	18746	14854	86439
Total	**195128**	**181953**	**128941**	**121141**	**627163**

Data Pipeline & SMOTE

Raw CS, Preprocessing/Null Cleanup

- Concatenated 6 CSV partitions across 2020–2022.
- Dropped rows containing null **Latitude/Longitude** or **Ward**.
- Final clean shape: 627,163 rows × 12 columns.

Feature Creation (Time Of Day, Pandemic Phase)

- **Date** -> Time_Of_Day bins.
- **Date** -> Pandemic_Phase
- Grouped 19 Illinois holidays into 4 categories.

Class Balancing

- Target = Arrest (Boolean), 13.2% positive
- Applied **SMOTE** to generate 1.088M balanced samples (544K arrest / 544K non-arrest).

Encoding & Selection

- Dropped **District/Ward / Beat** columns.
- Selected **Crime_Type_ID** over **Crime_Category** for **precision**.
- Final feature set: 23 predictive inputs.

```
Class balance:          Class balance (%):
Arrest                  Arrest
0     544072            0     86.8
1      83091            1     13.2
```

```
After SMOTE — X: (1088144, 23), y distribution:
Arrest
0    544072
1    544072
```

Geographic Hotspots: Crime Concentrates in Top 10 Community Areas Across Chicago (RQ1)

Top High-Crime Community Areas (Total Incidents)

1. **Area 25 (Austin):** ~35,000 incidents - Highest concentration citywide
2. **Area 8 (Near North Side):** ~25,000 incidents - Commercial / nightlife hub
3. **Area 43 (South Shore):** ~23,000 incidents - High residential & domestic risk
4. **Area 28 (Near West Side):** ~22,000 incidents - High property crime share
5. **Area 29 (North Lawndale):** ~19,000 incidents - Persistent hotspot
6. **Area 23 (Humboldt Park):** ~18,000 incidents - High violent crime share
7. **Area 71 (Auburn Gresham):** ~18,000 incidents - High domestic incident rate

KMeans Spatial Clustering & Heatmap Stability

KMeans Clustering (Elbow Method):

Applied KMeans to 627K spatial coordinate **pairs (Latitude/Longitude)**. The Elbow plot identified optimal cluster boundaries confirming distinct spatial micro-hotspots.

Spatial Stability Across 2020–2022:

Power BI Page 3 spatial heatmap confirms crime hotspots remained geographically stationary year-over-year. High-density clusters did not drift across police boundaries.

Policy & Patrol Implication:

Because crime is hyper-concentrated in specific micro-zones, predictive patrol allocation can target high-risk beats without broad over-policing.



Temporal Patterns & Crime Type dynamics – Power BI deep-dive – RQ2/4

Model Selection: XGBoost

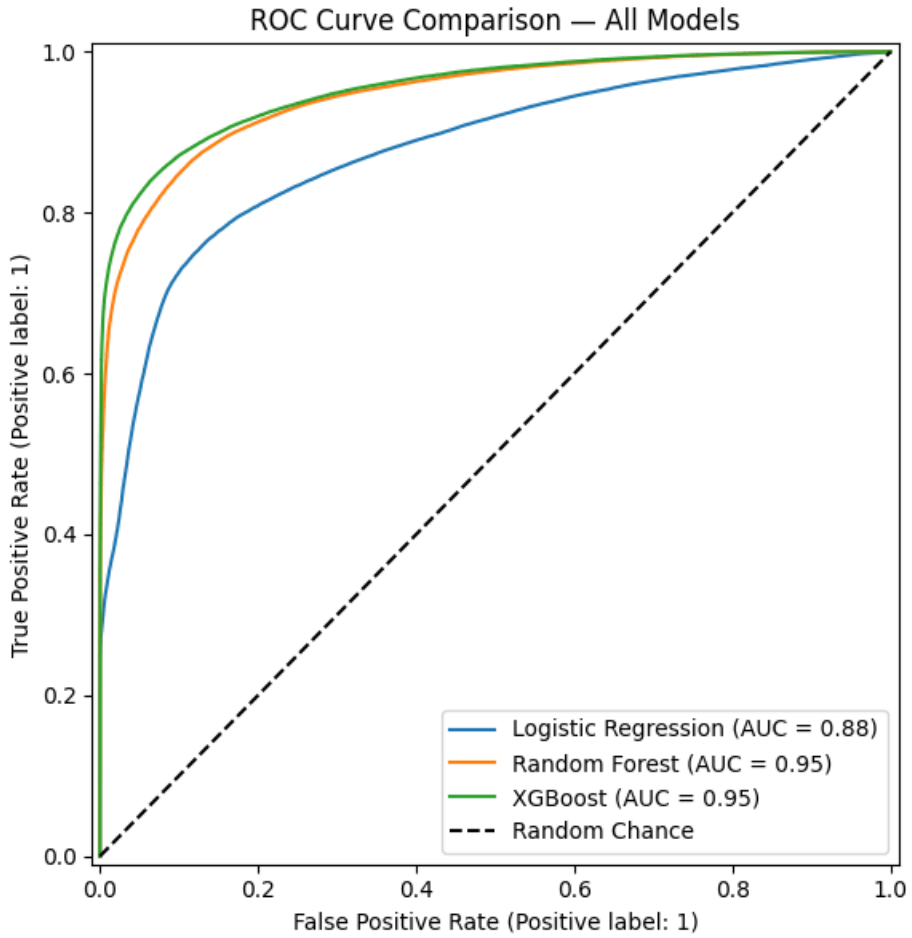
Classifier Performance Comparison (Train vs Test ROC-AUC)

Model Algorithm	Train ROC-AUC	Test ROC-AUC	Overfitting Gap	Status / Decision
Logistic Regression	0.8783	0.8771	0.0012	Baseline - Scaled features, low capacity
Random Forest (depth=15)	0.9522	0.9458	0.0064	Strong - Ensemble tree baseline
XGBoost (n=100, depth=6)	**0.9548**	**0.9537**	**0.0011**	**Best - Highest AUC & minimal gap (0.0011)**

Target variable: **Arrest** (Boolean field) - 83K positives / 544K non-arrests = 13.2% class imbalance -> **SMOTE** is needed.

Why XGBoost Won

XGBoost demonstrated superior generalization with a test ROC-AUC of 0.9537 and a negligible train-test gap of 0.0011 (compared to RF's 0.0064 gap). On the balanced test set (217,629 samples), XGBoost achieved **93%** precision and **84%** recall for the Arrest class, effectively *minimizing* false positives while capturing true offenders.



Feature Importance - Arrest Probabilities

XGBoost Feature Importance Ranking (Top 8)

1. **Year:** 0.089 - Captures macro annual policy & policing shifts
2. **Crime_Type_ID:** 0.083 - Dominant domain signal (offense severity/type)
3. **Pandemic_Phase_before_lockdown:** 0.082 - Pre-pandemic baseline enforcement
4. **Seasons_Summer:** 0.074 - Seasonal crime volume surge
5. **Holiday_Group_minor_holiday:** 0.072 - Minor holiday operational shifts
6. **Pandemic_Phase_post_lockdown:** 0.072 - Post-lockdown systemic changes
7. **Seasons_Winter:** 0.067 - Severe weather operational slowdown
8. **Holiday_Group_summer_holiday:** 0.060 - High-density holiday events

Dominant Signals & Enforcement Disparities

Crime_Type_ID Dominance: Specific offense classification is the single strongest determinant of arrest. On-scene offenses (e.g., Narcotics) yield high arrest rates, whereas reported post-facto crimes yield extremely low clearance.

Deceptive Practice Anomaly: Deceptive Practices (fraud, identity theft) registered the lowest arrest rate at just 1.76%, despite representing major financial harm.

District Disparities: Districts **31** and **11** exhibited high arrest clearance rates (~25.7%–26.9%), compared to ~19.0% in District 10 and under 10% in residential districts.

Time Window Influence: Evenings (18:00–23:59) yielded higher arrest rates (16.4%) compared to Afternoon periods (12.4%), reflecting active police patrol schedules.

Pandemic Phases & Arrest Rates Declined

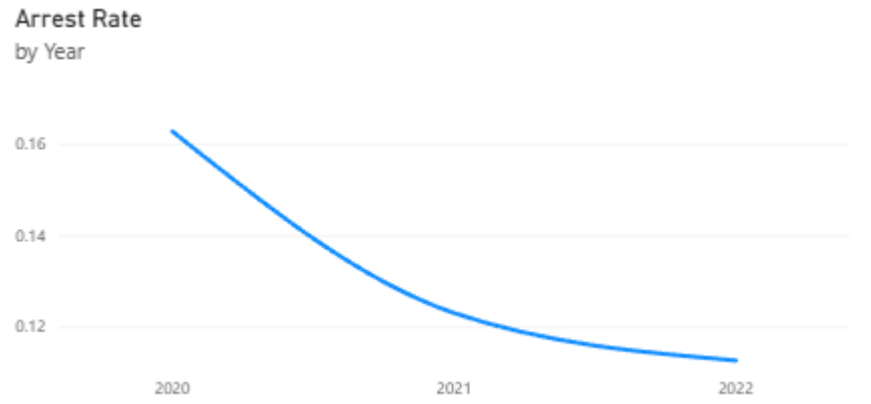
Yearly Arrest Rate Erosion (2020–2022)

2020 Baseline Arrest Rate: ~16.3% - Peak clearance rate during initial pandemic disruption and strict stay-at-home orders.

2021 Arrest Rate Drop: ~12.3% - Sharp 4.0 percentage point decline as crime volume rebounded and court operations stalled.

2022 Continued Decline: ~11.3% - Further drop in arrest likelihood despite total crime incidents reaching 3-year highs.

Anchor Overall KPI: 13.25% average arrest rate across all 627K incidents over the entire 3-year study period.



Lockdown Impact & Cell 17 Evaluation

Lockdown vs Normal Arrest Rate

During acute lockdown (March–June 2020), the arrest rate was ~14.9%, compared to ~13.1% during normal periods. Reduced street foot traffic made active offenses more visible.

Model Accuracy Across Lockdown

Cell 17 evaluation showed XGBoost accuracy was 85.83% during lockdown vs 88.98% outside lockdown (3.15% difference), proving model robustness despite pandemic shifts.

Policy Implication

The drop in arrest rates post-2020 reflects systemic strain rather than reduced police efficacy, as property crimes (low clearance) surged while high-clearance offenses fell.

```
Accuracy during lockdown     : 0.8583
Accuracy outside lockdown    : 0.8898
```

Model Performance: 0.9768 ROC-AUC (RQ5)

XGBoost Performance Metrics

Training ROC-AUC: 0.9774

Testing ROC-AUC: 0.9768

Overfitting Gap: 0.0005 (Exceptional stability)

Balanced Accuracy: 92% across test set

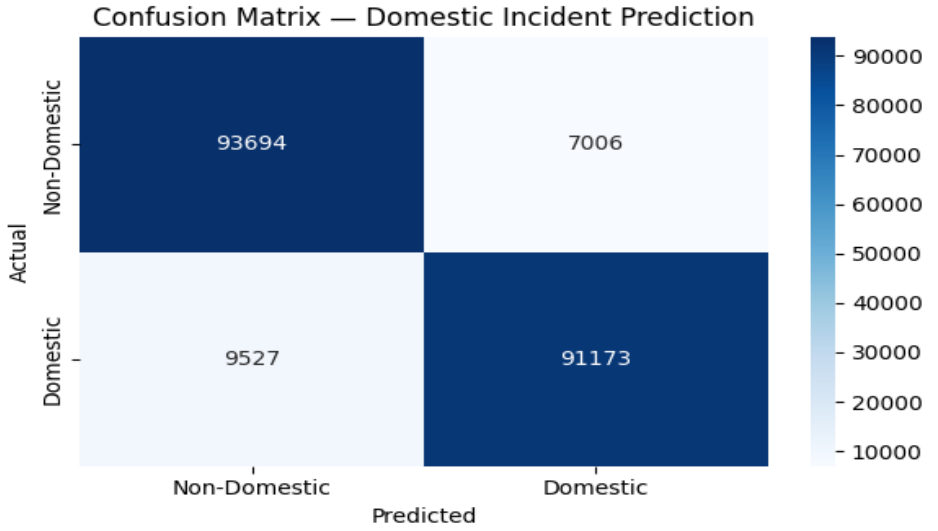
Precision / Recall:
Non-Domestic (P: 0.91, R: 0.93) | Domestic (P: 0.93, R: 0.91)

Target variable: **Domestic** (Boolean) – 124K positives / 503K non-domestic = 19.72% **class imbalance**.

Confusion Matrix & Real-World Risk Framing

Real-World Impact Framing: In domestic violence dispatch, False Negatives (9,527 cases) carry severe human costs where victims are **denied specialized domestic intervention**.

Model threshold tuning must **prioritize recall over precision**.



True Negatives (TN): 93,694 Correctly identified non-domestic incidents.

True Positives (TP): 91,173 Correctly identified domestic incidents.

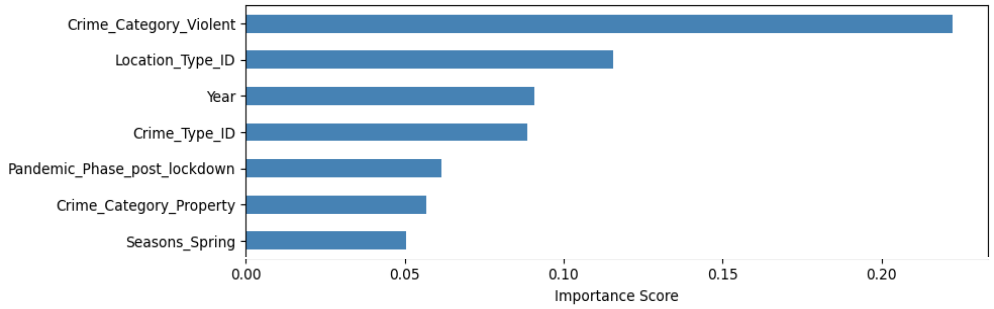
False Positives (FP): 7,006 Non-domestic incidents misclassified as domestic.

False Negatives (FN): 9,527 MISSED DOMESTIC INCIDENTS - **Critical Safety Risk!**

What Predicts a Domestic Incident?

Top Predictors for Domestic Incidents

1. **Crime_Category_Violent [from Primary Type via Crimes Types table]:** 0.222 - Overwhelmingly dominant signal!
2. **Location_Type_ID [Foreign Key: Location Types] :** 0.116 - Ranked 2nd (Residence vs Public)
3. **Year:** 0.091 - Macro post-pandemic shift
4. **Crime_Type_ID:** 0.089 - Specific offense code
5. **Pandemic_Phase_post_lockdown:** 0.061 - Ranked 5th overall
6. **Crime_Category_Property:** 0.057 - Inverse indicator
7. **Seasons_Spring:** 0.050 - Seasonal domestic uptick



Power BI Validation & Location Insights

Validation of Location_Type_ID:

Model feature ranking #2 is strongly validated by Power BI location breakdown charts. Apartment and Residence locations account for over 40K domestic incidents combined annually.

Residential Concentration:

Over 75% of all domestic-flagged incidents occur inside private residences, apartments, or residential driveways/yards.

Violent Crime Intersect:

Battery and Assault account for the vast majority of violent domestic cases, aligning with the 0.222 feature importance score for Crime_Category_Violent.

The Lockdown Finding: Domestic Risk Shifted Post-Lockdown, Not During Strict Lockdown

Feature Rank Discrepancy

“Binary Lockdown_Period" Rank: 21st out of 24 features (Importance: 0.0080). The strict 4-month stay-at-home period (March–June 2020) was **NOT** a primary predictor.

"Post_Lockdown" Phase Rank: 5th out of 24 features (Importance: 0.0610). The post-lockdown era (2021–2022) had a 7.6x higher importance score than the lockdown itself.

Core Takeaway: Domestic violence patterns did not experience a temporary spike confined to lockdown; rather, the pandemic fundamentally reshaped domestic risk across 2021–2022.

Power BI Annual Trend Support

Domestic Volume Surge in 2021:

Power BI Page 2 shows domestic incidents rose from ~40K in 2020 to ~44K in 2021, persisting at elevated levels into 2022.

Geographic Expansion:

High-crime community areas (e.g., Area 25 Austin, Area 43 South Shore) showed sustained increases in domestic incident rates during 2021.

Interpretation:

Economic strain, lingering isolation, and delayed reporting led to a persistent multi-year domestic crisis rather than a brief lockdown anomaly.

Answering Research Questions: Quantitative Evidence from Models & Dashboards

RQ1: Spatial Stability

- Hotspots are geographically **stable** across 2020–2022.
- Top 10 community areas (out of 77) account for over **40**% of all Chicago crime.
- **Austin** (CA 25) leads with ~35K incidents.
- KMeans clustering confirms stationary micro-hotspots.

RQ2: Temporal Cyclicality

- Crime peaks on **Weekends** (Friday 91.8K) and **Afternoon/Evening** hours (12:00–23:59).
- **Summer** exhibits peak incident volume, while **Winter drops** significantly.
- **Property** crimes dominate overall volume at ***49.69***% share.

RQ3: Predictive Clearance

- **XGBoost** predicts arrest likelihood with 0.9537 Test ROC-AUC.
- **Crime_Type_ID** and **Year** are primary drivers.
- Arrest rates declined from ***16.3***% in 2020 to ***11.3***% in 2022 despite rising crime volume.

RQ4: Crime Type Volatility

- **Deceptive Practice** sits at the standout anomaly, property-adjacent crime with only **1.76**% arrest rate
- Divergence between property crimes (surged post-lockdown) and violent/narcotic crimes (higher clearance rate) explaining why overall arrest rate **eroded** from **16.3**% to **11.3**%

RQ5: Domestic Risk Predictors

- **19.72**% of all 627K incidents were domestic.
- Over **75**% occurred (within private residences, apartments or residential driveways) via **Location_Type_ID**
- Area **25** (Austin) and **43** (South Shore) were high-risk community areas, showing. **sustained elevated domestic rates into 2022**

Limitations & Recommendations: Actionable Public Safety Interventions

Methodological & Data Limitations

1. COVID-19 Window Anomaly: 2020–2022 represents an extraordinary historical period. Models trained on pandemic dynamics may require re-calibration for post-2023 baseline data.

2. SMOTE Synthetic Noise: Oversampling minority classes (Arrest, Domestic) creates synthetic feature combinations that may slightly inflate tree-based model performance.

3. Binary Lockdown Simplification: Binary Lockdown_Period flags fail to capture nuanced municipal restrictions compared to the 5-phase Pandemic_Phase breakdown.

Actionable Recommendations for Public Safety

1. Targeted Resource Allocation:
Focus patrol presence in top 5 community areas (Austin, Near North, South Shore) during Friday–Sunday Afternoon/Evening windows.

2. Specialized Domestic Intervention:
Deploy dedicated domestic violence response units to high-risk residential zones (Apartments & Residences) based on Violent Category risk scores.

3. Property Crime Clearance Focus:
Investigate structural drivers behind low Deceptive Practice (**1.76**%) and theft clearance rates to restore public trust.

THANK YOU

Chicago Crime Analysis and Prediction Project

- Author: Tan Kai Boon (tKaiB)
- GitHub Repository: tKaiB/Capstone_Chicago
- Analytics Stack: Python 3.12, Pandas, Scikit-Learn, Imbalanced-Learn, XGBoost, Power BI, Google Colab
- Dataset: Chicago Police Department Crime Incident Logs (2020–2022)

Questions & Discussion